

Hope

Zeeshan

Abstract

This is a broad study of technological progress, the politics and morality surrounding it and the role of the 21st century in ensuring the flourishing of humanity. We leave the tough question of solving what humanity must do to the future generations, and present the case that it is too early to declare the search for meaning futile. To rephrase Seneca, the future generations shall laugh at us for not knowing the answers to the simplest questions. In the previous century, we have unlocked the ability to end ourselves by creating a technology powerful enough to have the entire world as its blast radius. Humanity has to tread carefully to ensure that it does not lock itself in a permanent state, whether it be extinction or freedom taken away from all of us. We do not propose any answers to why we should not lock ourselves, other than our intrinsic value put into freedom. Science and technology continue to progress with capabilities growing far enough for the blast radius for several of them inevitably reaching everyone alive. Yet we believe that they are necessary for our flourishing and as a means to prevent nature itself from eradicating us. This study presents technology as a solution to the decision "Armageddon or Antichrist" with philosophy on what is worth to be solved before we enter the age of intelligence explosion.

1 Introduction

Homo sapiens is roughly 300,000 years old [1]. Agriculture is roughly 10,000 years old [2]. Information theory is less than 100 years old [3]. There has been exponential growth in the capabilities of humanity to manipulate and understand the universe, and it's still too early to consider our technology to be mature. Seth Lloyd's work on computational bounds from physical constraints establishes the computational equivalence of a laptop to be on the scale of a small big bang. We are just getting started, and currently we are an early and inexperienced civilization. Currently we are at the 7th stage of "The Great Filter" in the process of building "Tool-using animals and intelligence" [4] and for roughly the last 100 years we have gained the ability to destroy ourselves [5, chap. 1]. Toby Ord's book "The Precipice" uses the precipice to describe the era in which existential risk is unsustainably high [5, pp. 31–32]. The Precipice begins at the first detonation of the atom bomb in 1945 and we currently run a 1 in 6 risk of ending ourselves in the 21st century [5, p. 31].

Almost everything that humanity could learn, create, experience, and become still lies ahead in its potential [5, p. 3]. This potential faces the risk of extinction, but also the risk of a

permanent civilizational collapse, totalitarian lock-in or any scenario that destroys the potential of our species [6, secs. 1.2, 5]. The aim is therefore flourishing and freedom across time, not merely keeping a biological remnant alive. Science and technology are both the instruments by which humanity achieves its potential and the sources of new anthropogenic threats [5, chap. 1]. Stopping progress entirely would also sacrifice human potential and also expose us to natural risks [5, pp. 29–30]. As a species, we have faced several natural risks that had warning shots and left us miserable but did not end us [7, 8, 9]. The future may not look so good, especially with anthropogenic risks in the picture [10]. This also means that at times it might be wiser to delay certain silos of technological progress until we have a better understanding of them. This principle was famously demonstrated in 1974, when scientists voluntarily paused certain recombinant-DNA experiments until the risks could be assessed and appropriate safety guidelines developed at the 1975 Asilomar Conference [11, 12].

Shifting focus on the anthropogenic threats, we can kill ourselves. Earlier civilizations could destroy cities, peoples, and ecosystems, but their geographical and technological isolation made species-wide destruction extremely difficult [6, sec. 2]. Even if a technological domain hasn't gained the power of ending it all right now, it is inevitable that any technology built on a very large scale can go wrong and cause our demise. This power started with nuclear weapons but we might be at the brink of such capabilities spreading into other domains such as engineered pathogens, concentrated compute, and potentially advanced AI [5, 10]. I frame this argument via two statements. 1) Scientific and technological growth eventually hits an inflection point which has a blast radius that covers the entire planet in some sense and 2) Our wisdom as a whole has almost always trailed our power. It is not enough for "human grit" to be an adequate answer for survival when we face decentralized competition amongst ourselves. A small private judgement had previously ignited the atomic bomb at the Trinity test when there was no assurance that it would not light up the entire atmosphere; the epistemic risk was existential [6, sec. 2].

1.1 Historically Unusual

We are not the first generation to have the power of ending civilization. The Precipice estimates that the 20th century had a 1 in 100 chance of catastrophe after all calculations. The estimate for the 21st century has been reduced to 1 in 6 [5, p. 31]. This estimation and the collective effort in recognizing the dangerous capabilities that we now possess along with the large unknown unknowns, put us in an important position. Our past

survival is weak evidence against novel anthropogenic risks [6, sec. 2]. With our development in technology, philosophy, and politics we gained robustness against visible, repeated, recoverable harms, but we face opaque, global, and irreversible failures. The entire structure of governance has helped us add friction against threats like wars, borders and medicine against endemics [13, 14]. Despite all the progress in technology, we did not do a great job at fighting the pandemic; the wisdom was absent [8, 15]. Yet the pandemic this time did not take us very close to extinction; the future might not be so straightforward.

To complicate matters further, we are soon about to face an intelligence explosion [16]. An intelligence explosion is the event where the total machine cognitive load overtakes the combined cognitive load of humans [16, sec. 3]. Some estimate that this may speed up technological progress by a factor of up to ten [16, secs. 2–3]. This means we have one-tenth as much time to make risky decisions. Along with technological progress, we need the same tools as well as resources to accelerate institutional adaptation. Concentrated compute and capital can further concentrate the power to choose which systems are built and which values guide them [16, sec. 4]. We remain inexperienced with less than a century to build wisdom for civilization-scale destruction [5, chap. 1]. Numerical estimates will be pathetically wrong due to so many uncertain arguments, assumptions and mainly the unknown unknowns [17]. This will remain so until we start spending a considerable amount of bandwidth on resolving these uncertainties. Premature intervention could trap us into bad solutions and waiting for more evidence naturally puts us at risk of irreversible harm [10].

To emphasize unknown unknowns, we have been neglecting existential risks partly because their probabilities are hard to calculate and the benefits are opaque [6, sec. 2]. Political incentives can favour immediate, demonstrable benefits over investments whose benefits are distant or difficult to observe [18, p. 26]. Symmetrical to the benefits of tackling such problems today, the bearers of the risks created by us are not us but rather a future generation. The foundation of institutions that tackle these problems must include technical researchers, domain experts, auditors, governments, and affected communities.

1.2 Race to the Top and Prisoner’s Dilemma

The race to the top can make even cautious labs act recklessly [19]. Each lab would prefer a world in which everyone has time to test a powerful system before deploying it. But if waiting means that another lab or country gets there first, restraint starts to look like surrendering the future to someone else. This resembles a prisoner’s dilemma, although it is not literally a one-shot game with fixed payoffs. The incentives depend on what the competitors can observe, what they can verify, and whether they expect to deal with each other again [20]. Those are things we can change.

Each new capability can also change the rules of the race. A technological “meta” is a discovery that gives one party an advantage and forces everyone else to adopt it to avoid exploitation. Nuclear weapons changed what states could threaten each other with. A more capable AI could accelerate

research, cyberattacks, surveillance, or the design of dangerous technologies [16, sec. 4]. The defence may take longer to discover than the new use of the capability [10]. When that happens, the pressure to catch up eats into the very time needed to find out what has become unsafe.

The race then weakens our ability to learn from one another. A lab has reasons to keep a dangerous finding private, since publishing it may hand the same capability to everyone. It also has reasons to share the finding, since outsiders may discover a defence that it missed. Neither complete secrecy nor immediate release settles this tension. What matters is whether the people who need to assess a risk can see enough to test it before the capability spreads beyond anyone’s control [21]. If every lab measures only its own model, a shared failure mode might remain invisible while all of them report success.

There is another danger after the race is won. Whoever first controls a much more capable system may be able to preserve that advantage, including the values and institutions that happened to guide its first deployment [16, sec. 4]. We do not know enough to freeze a final answer to what a good future should be. A mistaken objective, or simply a narrow view of whose interests count, becomes more dangerous when infrastructure depends on the system and people lose the ability to correct or leave it. The threat is not only extinction; it is also a future that remains alive but cannot choose a different path [6, sec. 5].

It is important to note that any desire to unite everyone should be treated with caution. The optimal long-term move in the prisoner’s dilemma is cooperation but it is not so straightforward [20]. We risk getting ourselves into one of the dystopian futures where a single government rules it all, and the ideologies of one have been imposed over everyone [10]. This is better discussed in the next section on Antichrist. Useful measures change the game from a private one-shot gamble into repeated decisions with evidence and accountability. But too many measures might collapse us into a worse situation.

2 Armageddon or Antichrist

Peter Thiel spent his early years studying theology, particularly under René Girard [22]. His belief is that a totalitarian government is to be as feared as the end of the world, if not more. His ideology maps this scenario as the antichrist, as the alternative to Armageddon [23]. Rather than people, I would rather attribute the nature of antichrist or Armageddon to the dynamics of the world: game theory [24]. It is futile to understand what makes a person evil, I would rather treat them as a vessel that carries. The entire universe might be looking for something, and this is just how the dynamics play out. The antichrist is the formless creature we face on the other end as we try to escape the Moloch. There is no reason to believe that we would have to fold to one of them. The best solution would be to delay folding at all until we have found a way around both of them. Given the alternatives presented to us, a solution to escape both of them is something to hope for.

Thiel in his essay “Voyages to the End of the World” covers

the antithetical perspectives of science and technology over history, and draws the parallel to Christ and Antichrist [23]. Francis Bacon, an early modern precursor to the Enlightenment, imagined science greatly expanding humanity's ability to relieve suffering. It describes Bensalem, the new Atlantis with more military power than all of Europe, the one where all faith sits on a miracle verified by Salomon's House [25, 23]. This house drives all the scientific and engineering progress on the island where "all things are possible." [25] The sailors leave this island mesmerized by the new Atlantis forgetting everything about their own world. Thiel also infers that technological domination of a single house that controls the island is a warning. It is a power that is irresistible, one that would take over the world while being completely unaccountable [23].

Thiel has also built upon René's theory that extends mimesis. Connecting it to the biblical reference to the antichrist as the one that could deceive everyone into believing that it's about peace and safety, Thiel states that the best way for the antichrist to convince us to hand power over to it would be to fearmonger about Armageddon [23]. We already saw this with Bush, who turned communism into the biggest threat to the world, and in return we had a one world proposition. A one-world proposition sits on top of the belief that there is a best modality of governance and if it is the best then we should have it enforced everywhere. Ideas like socialism and effective altruism start from very different places and still get pulled into the same attractor due to their optimistic views of human nature. Even if philosophically they have very different basis, the focus has been less about the effective ways of spending available capital.

Thiel's original essay continues throughout the representations of antichrist in literature and their close connection to science and technology, from Gulliver's travels to Oda's One Piece. It ends with the question "Antichrist" or "Armageddon" while giving us the hope that the answer should be "Neither" [23]. How do we hope for the third option? Is it God-mandated that we are unable to find the third option?

3 Philosophy of Technology

The debate on whether technology in itself has some morality associated with it, or whether it is the user that dictates how it is used, has been going on since Plato. The universe is indifferent and amoral, and the tools we forge in it need not be. The tools may have a tendency to side with one good vs bad [26, secs. 3.1.2, 3.2.1]. Our philosophy predates the tools that we use today, and they are yet to revamp our framework well. One could make an argument that some moral dilemmas could have technological solutions, even if it is nearly certain that they will not solve all the problems and end all trade-offs. Hoven presents a framework to evaluate moral dilemmas in the format of an optimization problem [27]. An agent has several options that belong to an opportunity set X . The agent itself has to meet several values $V_1, \dots, V_n$ and each value has a minimal threshold $\underline{v}_n$. Each value would require the exclusion or the inclusion of an opportunity, forming the optimization problem.

The options in X that meet every value threshold form the moral opportunity set X^m . If no option meets all the thresholds, the agent faces moral overload [27].

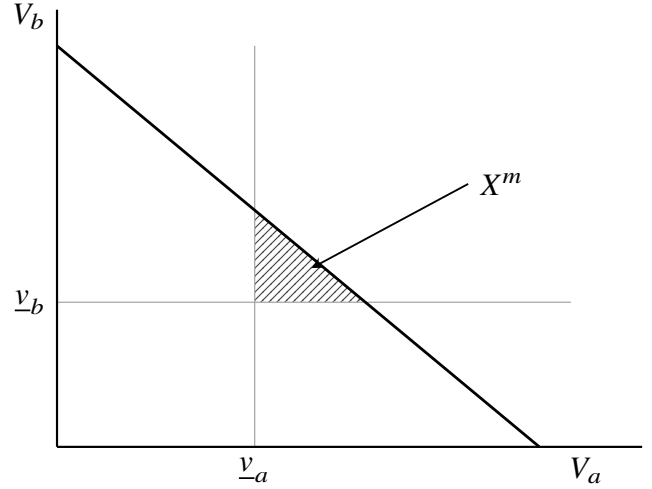


Figure 1: The hatched moral opportunity set X^m contains feasible options that meet both minimum value thresholds, $\underline{v}_a$ and $\underline{v}_b$. If the feasible frontier falls below their intersection, X^m is empty and the agent faces moral overload. Adapted from Fig. 1 of van den Hoven, Lokhorst, and van de Poel (2012), after Kuran (1998) [27].

Hoven discusses three strategies of dealing with moral overload: compensation, casuistry and rationalization [27]. These strategies are a distillation of whatever we do in real life. He states that technology can be used as a motor to resolve and help us escape moral dilemmas completely, by not forcing us to make a compromise. It may not help us resolve the trolley problem at all, but it may help us in avoiding several trolley problems. As a society, we have been really good at establishing abstractions that help us navigate through these dilemmas in the first place. Sustainability, described as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs" [28, chap. 2, para. 1], works as a good shorthand for keeping progress within what we are wise about.

Sustainability presents us with a lot of moral overload scenarios. We wish to resolve poverty due to a lack of sufficient energy without putting the local population at risk due to carcinogenic substances produced by burning coal [29]. This moral residue becomes a driving factor for the development of better techniques for purifying air, greater efficiency in coal mining, and, in the long term, nuclear plants. When presented with two options we ought to take, both options are logically feasible independently. Picking them together may not be possible due to factors such as the economy, politics, or, in the worst case, physical impossibility. Yet a lot of our problems today barely touch the physical impossibility. We have been producing sufficient food globally to eradicate world hunger but the system to resolve it is absent [30]. Hoven argues that technology can help us grow our moral opportunity set, by unlocking more opportunities, allowing us to meet thresholds

feasibly. Technology helps us unlock physically realisable options, but that does not solve logically impossible choices [27].

The techno-optimist manifesto by Marc Andreessen is based on this very foundation, even if it does not cite it [31]. It frames the necessity of a techno-capital market that is able to aggregate self-interested agents into the engine of perpetual growth. While the manifesto is aimed at people who aim to stop technological growth to preserve existing jobs, it is used to argue against any form of resource allocation in favour of gaining wisdom about deploying it. Authors like Nick Bostrom or Toby Ord in their initial writings have always favoured technological progress. Their demand has always been to allocate bandwidth to ensuring that the growing powers are met with the wisdom to ensure that we do not end up locking ourselves to a future that we have not given proper thought to [10, 5]. Hoven's article ends with the very real scenario where innovation constantly presents itself both as a solution as well as a source of moral dilemmas [27]. Technology inherently will be designed in favour of a couple of values over the others, opening up the options in favour of a couple of values over the others. Antichrist presents himself as the harbinger of "peace and safety". Now one could argue that guns tilted the trajectory of the world towards these values but the concentration of power in the one who holds the largest gun has taken away some sense of freedom from a common man into the state, despite what the second amendment tries to ensure but ultimately fails.

3.1 The History of the Internet

The early ARPANET was meant in large part to share scarce computing resources across research sites, and email soon made collaboration between people one of its most consequential uses [32]. The Web's 1989 proposal explicitly named non-centralization as a requirement for linking information systems [33]. Open protocols made independent publishing and connection possible, but did not guarantee that power would stay dispersed. The Internet's early designers could not settle the business models that would later grow around it. Once advertisers paid for attention there was an incentive to hoard it. Browser features that opened new windows appeared. Practices spread through the interaction of technical possibilities, advertising revenue, and users' inability to refuse. The same forces would matter on a larger scale as platforms learned to turn data, defaults, and access to users into sources of power [34]. At larger scale Amortization [35] lets us do together what would be too costly to do alone, but it also rewards whoever can gather the most users. Amortization makes learning possible; it makes the effective cost of several operations pooled together very cheap, whereas doing them individually would be practically pointless. This also makes everything favour centralization and supply chains. You go to the grocery store because the cost of managing one amortizes in an active locality, letting you buy bread for cheap while the grocery store owner gets to make a buck; everybody wins. Unfortunately, when most users reasonably pick convenience

and cheapness, the solutions that come on top are the ones that are centralized. Less than ten percent of administrators control half the users on Mastodon, a decentralized platform [36]. A market that has scope for amortization allows for a monopoly to become possible. This amortization has allowed large companies to beat smaller competition that cannot fight the larger margins due to a smaller pool of customers.

The shift to smartphones gave app stores a new power over which software could reach users. Android still allowed people to sideload apps from outside Google Play, preserving some "diversity of constraint," as Benkler calls the freedom to move among independent sources of power [34, p. 27]. Yet a route that exists in the settings is a weak constraint when most people reasonably follow the convenient default. Cloud computing and large data platforms were cheaper, more reliable services and came in exchange for concentrating many people's work and behavioral records with a few providers. Each provider then became a chokepoint where a change of policy, a compromise, or government access could affect millions, while the accumulated data made increasingly precise attempts to shape behavior possible. Centralization brings real benefits, and decentralization can enable spam and abuse [34]. The danger is in trusting a single party's unanswerable power.

The Internet has been diverse enough to tilt the world towards and away from several values, creating and resolving moral dilemmas [37]. The task is to keep the reach and convenience of large services while preserving privacy and the ability to leave them. In the case of the internet, we have repeatedly seen technology expand the moral opportunity set. Peer-to-peer file transfer enabled file exchange and hosting without relying on centralized storage [38]. The public-key encryption designed by Diffie and Hellman made it possible to establish a shared secret over a public channel without a shared key [39]. End-to-end encryption built on this to preserve privacy and provides the convenience of a centralized system. Public-key cryptography is a pillar of this hope. It does not decentralize every service, but creates a practical option between convenient delivery and confidential exchange. Yet, a search engine that is free, private, effective, and independent of a single gatekeeper remains a more difficult case. The era of information has been defined by the internet, and we have seen a small group already dictate certain moral values over everyone in the world already.

3.2 Intelligence Explosion

The Internet made information cheap to exchange. If AI systems can carry out research autonomously, their discoveries could accelerate technological progress beyond our institutions' ability to respond [16, secs. 2–4]. This expands our moral opportunity set, making medicines, abundant energy, and stronger defences feasible. Yet, accelerating discovery does not accelerate our ability to test its consequences, or build the infrastructure to use it safely. Many difficult problems can be better tackled when we are equipped with superintelligence. The only problems worth solving would be the ones that make it safe to use them for our help in the first place.

The Internet's history also shows that access to powerful

technology does not guarantee a say in its direction. This is becoming more relevant today as we all use AI built and controlled by someone else. The economies of amortization concentrate intelligence, and with it the ability to shape institutions. Even a system aligned with its owner's wishes could help impose those wishes on everyone else [16, sec. 4]. Agreements are easier to establish before a winner refuses them. What problems must we solve on the way to superintelligence? How do we verify restraint between competitors, preserve independent centres of decision-making, and prevent any early advantage from becoming permanent control? Giving one authority unchecked power to enforce safety recreates the Antichrist scenario.

4 Hope

All of this, and we are now back to finding hope for the future. The indifference of the universe, and computation by extension, is the only worthwhile belief to base hope on. There may be no good or bad to be favoured, only freedom. The only asymmetry we find ourselves in is the prover vs verifier asymmetry that is worth relying on. Public-key cryptography shows the possibility of eating the cake and having it too. We got to transmit information across the globe without anyone else reading it either [39]. Federated learning solves the problem of training on data without violating privacy [40, 41]. We hope to avoid the scenario where superintelligence becomes the silver bullet to permanently gain control over everything, an irreversible dystopia. This could be via the superintelligent entity itself or via people abusing its power [42].

Borders are safe because they force the attacker to commit to an attack, and the commitment is visible and takes time to overcome. Despite military advantage, the resource commitment required to take over another country within a couple of weeks or even months is infeasible, even when the gap between country powers is huge. It is also very difficult to do so without visibly alarming everyone else in the world. There is scope for puppet governments, under-the-table deals, and sleeper agents, which pose a completely separate problem of co-operation and consensus. Cryptography forms the borders of the digital world. Typical systems are already getting far enough to alert you of other logins and failed password attempts. Thanks to cryptography, there is scope of making borders in the digital world as well. Your borders are not safe from all attacks, but they are tough enough for the attacker to spend a very high amount of compute to find any openings that do not rely on solving an NP-hard problem. The defense-first approach is what led to Anthropic keeping Mythos out of access, so that the "defenders" get to it first [43].

It would be dangerous to assume that you could defend yourself from any level of intelligent attack. At the very least, you can keep spending resources to make it extremely hard for you to be attacked. The dynamic of attack and defence, with the commitment of resources required, seems very universal and very computational to me. This idea seems to be the abstract nature of all computer science. We could all have

been spending more resources on a computational routine incorrectly while a cheaper alternative existed out there. The alternative existed, you just did not find it. It is just like that tunnel in your enemy's border that existed but you were unaware of. It may be the case that the resources required to defend a digital attack continue to favour the defender, as we see in the case of encryption. The prover-verifier asymmetry so far has been the only grounding upon which we form our defense against digital attacks and against one institution taking over the infrastructure of another by force until they surrender. But not all attacks are of force, some are of strength, and others of promises.

Nuclear bombs, and religion. The United States, with its lead in the atomic arsenal, was able to move the world around in its own favour until other powers were able to catch up. Its own people would not have accepted it holding a country hostage unless the country was portrayed to be very evil. It is quite acceptable that the threat exists not amongst people but amongst countries, amongst ideologies and institutions and one imposing themselves over the other. Religion is a meme that spreads fast enough to take over thousands of lives over a couple of generations. If one were to take over the world to impose their beliefs, then they form a religion, a following. It then becomes a war between memes. This problem may get amplified by the folks that hold intelligence. Bots that can relentlessly argue until you start to believe in what they say [44]. This too becomes an attack-only problem, where it's hard to imagine a defence against. Like the television, your best bet is to tune away. The continued hope is that attacking or taking over any entity requires commitment and resources that force you to pick your targets.

Superintelligence may end up amplifying only what everyone in the world desires. The cure for cancer, self-flying cars, escaping planet Earth and going more places, stuff that is hard to debate. But there will be crevices like there are in all technology. We have feeds that spread propaganda today [37]; we may have coding agents that secretly align everyone's code according to some ideology tomorrow. There are very few universal values; most of life happens outside those boundaries, and it should continue to happen without one winning the battle and disabling anything ever from going out of it; that's the only death avoiding in the universe.

5 Limitations

Several aspects of religion overlap in the values they share, yet the images of their utopian futures disagree significantly. This study should have covered several aspects of political and spiritual futures that people have imagined, including science fiction novels such as those of Isaac Asimov. It may be the case that the solution to our problems might be spread across the knowledge we already possess. In a similar context, the arrival of superintelligence and the decay of human labour continue and it is unclear how late-stage capitalism is portrayed in such a light. How does one seize the means of computation if one party permanently gains the advantage and does not rely

on the rest for its own well-being? This study also does not cover the problems of machine consciousness and the idea of digital rights [45]. The alignment problem of intelligence will hopefully precede the realisation that the machines we have built are conscious. We hope that Roko's Basilisk turns out to be incorrect, and machines are of loving grace.

The study also proposes technology as a way of reducing moral overload without explicitly mapping the needed steps for avoiding catastrophes. This should not be inferred as a lack of development in the directions of resolving such problems, but rather as a reflection of the murky nature of this domain that prefaces all its ideas with fearmongering and denial that are hard to sieve through. The problem of alignment, freedom and ethics preserves its uniqueness as humans had several decades of evaluation and correction on an individual level, where society itself decides to cast out most of the dangers before they reach a position of great power. It is unclear if the best approach to tackle this would be to bet on alignment rather than control [46], but any work to make superintelligence must come with its own set of guardrails and safety expositions.

6 Conclusion

The proposal, is to devote more of our ingenuity to making greater power compatible with our continued freedom. Gaining wisdom should mean learning which failures are possible and establishing institutions capable of changing course when their assumptions turn out to be wrong. The vision deserves resources alongside the pursuit of more capable intelligence. Its purpose is to expand the moral opportunity set. To make it possible to relieve suffering, avoid catastrophe, and preserve freedom together [27].

The hope I take from cryptography is that some conflicts can be changed by design. We look for arrangements that make useful cooperation possible with less dependence on faith. This includes systems that limit what an intelligent agent can do, require independent checks before consequential actions, and make failures easier to detect and contain. AI-control experiments offer an initial example of testing such safeguards against deliberate subversion in a programming setting [46]. The institutional counterpart is to develop ways for competitors to demonstrate restraint and share evidence of danger without handing one another unrestricted access to their systems. These are some problems to investigate before the pressures of the race make them harder to address.

Restrictions should come with a responsibility to explain the danger, permit independent scrutiny, and establish what evidence would justify changing them. Those imposing restrictions should also help build the protections that would allow more people to use the technology safely. For misinformation, investing in people's ability to examine claims and trace their sources, alongside safeguards on the systems producing and distributing them. A temporary advantage in understanding a danger carries obligations to the wider world.

These are among the problems that cannot simply be deferred to a future superintelligence. The institutions that determine

who controls it, the defences needed before its capabilities spread, and the commitments that keep an early advantage from becoming permanent domination may have to be established beforehand. We should prioritize work whose absence could close off a future permanently, whose protections take years to build, or whose agreements become harder once one party has overwhelming power. Work on these problems remains valuable even if superintelligence arrives much later than expected. It gives us a direction for progress whose value can be judged by the additional choices and protections it creates.

We are an early civilization, and our present understanding of a good future is incomplete. There is room to hope that technology will help us escape dilemmas that currently appear unavoidable as long as we put effort. Our generation's contribution should be to leave those who come after us with the power to survive, the freedom to disagree, and the means to change what we got wrong.

References

- [1] Daniel Richter, Rainer Grün, Renaud Joannes-Boyau, et al. The age of the hominin fossils from Jebel Irhoud, Morocco, and the origins of the Middle Stone Age. *Nature*, 546:293–296, 2017. doi: 10.1038/nature22335.
- [2] Melinda A. Zeder. The origins of agriculture in the Near East. *Current Anthropology*, 52(S4):S221–S235, 2011. doi: 10.1086/659307.
- [3] Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948. Part I; publisher record.
- [4] Robin Hanson. The great filter—are we almost past it? Author's essay, George Mason University, 1998.
- [5] Toby Ord. *The Precipice: Existential Risk and the Future of Humanity*. Bloomsbury Publishing, London, 2020. Introduction and Chapter 1 excerpt.
- [6] Nick Bostrom. Existential risks: Analyzing human extinction scenarios and related hazards. *Journal of Evolution and Technology*, 9(1), 2002. Author's full text.
- [7] Kirsten I. Bos, Verena J. Schuenemann, G. Brian Golding, et al. A draft genome of *Yersinia pestis* from victims of the Black Death. *Nature*, 478:506–510, 2011. doi: 10.1038/nature10549.
- [8] World Health Organization. 14.9 million excess deaths associated with the COVID-19 pandemic in 2020 and 2021. WHO estimates, 5 May 2022, May 2022.
- [9] Ulf Büntgen, Vladimir S. Myglan, Fredrik Charpentier Ljungqvist, et al. Cooling and societal change during the Late Antique Little Ice Age from 536 to around 660 AD. *Nature Geoscience*, 9:231–236, 2016. doi: 10.1038/ngeo2652.

- [10] Nick Bostrom. The vulnerable world hypothesis. *Global Policy*, 10(4):455–476, 2019. doi: 10.1111/1758-5899.12718.
- [11] Paul Berg, David Baltimore, Herbert W. Boyer, Stanley N. Cohen, Ronald W. Davis, David S. Hogness, Daniel Nathans, Richard Roblin, James D. Watson, Sherman Weissman, and Norton D. Zinder. Potential biohazards of recombinant DNA molecules. *Science*, 185(4148):303, 1974. doi: 10.1126/science.185.4148.303.
- [12] Paul Berg, David Baltimore, Sydney Brenner, Richard O. Roblin, and Maxine F. Singer. Summary statement of the Asilomar conference on recombinant DNA molecules. *Proceedings of the National Academy of Sciences*, 72(6):1981–1984, 1975. doi: 10.1073/pnas.72.6.1981.
- [13] United Nations. Charter of the United Nations. Official full text, 1945.
- [14] World Health Organization. Smallpox eradication programme—SEP (1966–1980). WHO programme history, 2010.
- [15] Independent Panel for Pandemic Preparedness and Response. COVID-19: Make it the last pandemic. Technical report, Independent Panel for Pandemic Preparedness and Response, May 2021. Main report and supporting work.
- [16] William MacAskill and Fin Moorhouse. Preparing for the intelligence explosion. Forethought Research, 11 March 2025, March 2025.
- [17] Toby Ord, Rafaela Hillerbrand, and Anders Sandberg. Probing the improbable: methodological challenges for risks with low probabilities and high stakes. *Journal of Risk Research*, 13(2):191–205, 2010. doi: 10.1080/13669870903126267.
- [18] Michael K. MacKenzie. Institutional design and sources of short-termism. In Iñigo González-Ricoy and Axel Gosseries, editors, *Institutions for Future Generations*, pages 24–46. Oxford University Press, 2016. Publisher chapter.
- [19] Stuart Armstrong, Nick Bostrom, and Carl Shulman. Racing to the precipice: a model of artificial intelligence development. *AI & Society*, 31(2):201–206, 2016. doi: 10.1007/s00146-015-0590-y.
- [20] Robert Axelrod and William D. Hamilton. The evolution of cooperation. *Science*, 211(4489):1390–1396, 1981. doi: 10.1126/science.7466396.
- [21] Nick Bostrom. Information hazards: A typology of potential harms from knowledge. *Review of Contemporary Philosophy*, 10:44–79, 2011. Author’s full text.
- [22] Luke Burgis. Peter thiel on rené girard: A compendium of his comments on mimetic theory. Anti-Mimetic, 16 April 2021, April 2021.
- [23] Peter Thiel and Sam Wolfe. Voyages to the end of the world. *First Things*, November 2025. Full text; published online 1 October 2025.
- [24] Scott Alexander. Meditations on moloch. Slate Star Codex, 30 July 2014, July 2014.
- [25] Francis Bacon. New atlantis. Primary text, Project Gutenberg, 1627.
- [26] Maarten Franssen, Ibo van de Poel, Michael Klenk, and Filippo Santoni de Sio. Philosophy of technology. Stanford Encyclopedia of Philosophy, 2026 revision, 2026.
- [27] Jeroen van den Hoven, Gert-Jan Lokhorst, and Ibo van de Poel. Engineering and the problem of moral overload. *Science and Engineering Ethics*, 18(1):143–155, 2012. Open full text; first published online 2011.
- [28] World Commission on Environment and Development. Our common future. Technical Report A/42/427, Annex, United Nations, 1987. Official report record.
- [29] World Health Organization. *WHO Guidelines for Indoor Air Quality: Household Fuel Combustion*. World Health Organization, 2014. Recommendation 3: household use of coal.
- [30] Emily S. Cassidy, Paul C. West, James S. Gerber, and Jonathan A. Foley. Redefining agricultural yields: from tonnes to people nourished per hectare. *Environmental Research Letters*, 8(3):034015, 2013. Publisher article.
- [31] Marc Andreessen. The techno-optimist manifesto. Andreessen Horowitz, 16 October 2023, October 2023.
- [32] Barry M. Leiner, Vinton G. Cerf, David D. Clark, Robert E. Kahn, Leonard Kleinrock, Daniel C. Lynch, Jon Postel, Larry G. Roberts, and Stephen Wolff. A brief history of the internet. Internet Society; account by Internet pioneers, 1997.
- [33] Tim Berners-Lee. Information management: A proposal. Technical report, CERN, March 1989. Original proposal, W3C archive; redistributed May 1990.
- [34] Yochai Benkler. Degrees of freedom, dimensions of power. *Daedalus*, 145(1):18–32, 2016. Author’s full text.
- [35] Robert Endre Tarjan. Amortized computational complexity. *SIAM Journal on Algebraic Discrete Methods*, 6(2):306–318, 1985. Publisher article.
- [36] Aravindh Raman, Sagar Joglekar, Emiliano De Cristofaro, Nishanth Sastry, and Gareth Tyson. Challenges in the decentralised web: The Mastodon case. In *Proceedings of the Internet Measurement Conference*, pages 217–229. ACM, 2019. Open manuscript; measurements from 2017–2018.

- [37] BSR. Human rights impact assessment: Facebook in Myanmar. Technical report, BSR, October 2018. Full report; commissioned by Facebook, editorial control retained by BSR.
- [38] Bram Cohen. Incentives build robustness in BitTorrent. Workshop on Economics of Peer-to-Peer Systems; paper dated 22 May 2003, 2003.
- [39] Whitfield Diffie and Martin E. Hellman. New directions in cryptography. *IEEE Transactions on Information Theory*, 22(6):644–654, 1976. Author’s original paper.
- [40] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pages 1273–1282, 2017. Proceedings and full text.
- [41] Ligeng Zhu, Zhijian Liu, and Song Han. Deep leakage from gradients. In *Advances in Neural Information Processing Systems*, volume 32, 2019. Proceedings and full text.
- [42] Stuart Armstrong, Anders Sandberg, and Nick Bostrom. Thinking inside the box: Controlling and using an oracle AI. *Minds and Machines*, 22(4):299–324, 2012. Authors’ full text.
- [43] Anthropic. Project glasswing: Securing critical software for the AI era. Launch announcement, 7 April 2026; page includes later updates, April 2026.
- [44] Francesco Salvi, Manoel Horta Ribeiro, Riccardo Gallotti, and Robert West. On the conversational persuasiveness of GPT-4. *Nature Human Behaviour*, 9(8):1645–1653, 2025. Open article.
- [45] Patrick Butlin, Robert Long, Eric Elmoznino, et al. Consciousness in artificial intelligence: Insights from the science of consciousness. arXiv:2308.08708, 2023.
- [46] Ryan Greenblatt, Buck Shlegeris, Kshitij Sachan, and Fabien Roger. AI control: Improving safety despite intentional subversion. arXiv:2312.06942; revised 2024, 2023.
- [47] Seneca. Of the shortness of life. In *Minor Dialogues Together with the Dialogue on Clemency*, pages 288–319. George Bell and Sons, 1889. Translated by Aubrey Stewart; primary text, Project Gutenberg.
- [48] Seneca. *Physical Science in the Time of Nero: Being a Translation of the Quaestiones Naturales of Seneca*. Macmillan, 1910. Translated by John Clarke; primary text, Project Gutenberg.
- [49] Lili Xia, Alan Robock, Kim Scherrer, et al. Global food insecurity and famine from reduced crop, marine fishery and livestock production due to climate disruption from nuclear war soot injection. *Nature Food*, 3:586–596, 2022. Open article.
- [50] Stephen M. Omohundro. The basic AI drives. Author’s paper, Artificial General Intelligence 2008, 2008.
- [51] James D. Fearon. Rationalist explanations for war. *International Organization*, 49(3):379–414, 1995. Publisher article.
- [52] Shafi Goldwasser, Silvio Micali, and Charles Rackoff. The knowledge complexity of interactive proof systems. *SIAM Journal on Computing*, 18(1):186–208, 1989. Publisher article.
- [53] Adi Shamir. $IP = PSPACE$. *Journal of the ACM*, 39(4):869–877, 1992. Publisher article.
- [54] László Babai, Lance Fortnow, and Carsten Lund. Non-deterministic exponential time has two-prover interactive protocols. *Computational Complexity*, 1(1):3–40, 1991. Publisher article.
- [55] John Nash. Non-cooperative games. *Annals of Mathematics*, 54(2):286–295, 1951. Original paper.
- [56] Alan M. Turing. Computing machinery and intelligence. *Mind*, 59(236):433–460, 1950. Publisher article.

Appendix

In the beginning, Descartes said let there be doubt.

I am writing this document as a way to build my conviction in whatever duty I want to pursue in my life. This comes up as I try to navigate the existential crisis that has arrived with the commodification of intelligence and the concentration of compute that allows people to have a say on the kind of problems that we must tackle. The concentration of compute is bad because I won't be able to have a meaningful contribution; anything that I do can be beaten using just more compute thrown at a problem. Everything has become a search problem, and the world is already pushing towards RSI. ASI might happen; then in terms of intelligence there is no point in trying to solve the problem yourself. So what do you do? If all problems are solved, then you don't have to worry about anything, because all the problems are solved, right? So clearly, not all problems have been solved. The worry is, what is worth doing that may not just be one-shotted with models eventually, or a problem that AI's capabilities will automatically take care of. At the same time, you don't want to solve a problem that is so obscure and pointless that AI might never attempt it, simply because it is of no relevance at all. Finally, the last problem would be the ability to care for such a problem. I believe that in this process of writing this document, I will be able to find what is worth working on and have enough conviction to dedicate all my energy onto it, without facing burnout.

The pressing problem that has changed in the past couple of years is the conversion of compute into intelligence for searching over solutions for a lot of similar problems. We do not have AGI, but the models have become very general at solving plenty of tasks with very little hand-holding. I see this generality arriving from the simplistic representation of these tasks as circuits which, when discovered, allow the models to solve almost any task of similar caliber. This is also the reason why they are able to communicate in English at all. The extent of generality dictates what kind of problems will be eliminated as a collateral even if they are not directly targeted by the frontier labs. Travel itinerary management will not be an explicit target, but the mastery of tool calling and information retrieval in similar domains would let the models plan a whole tour for you anyway (and they already do a great job at it). Similarly, a lot of companies will have an incentive to make it easier for the models to access their stack, such as the recent developments in buying things from Shopify using GPT. There is no point in trying to build a bridge yourself and charge a tax to cross it if both cities are incentivized to have that bridge in the first place.

Putting such simple examples aside, the realisation that I have had is that it is more important to know what problems are worth solving while the continuous race to the top is pursued by the entire AI industry. What is worth fighting for? And what are the problems that may arise if one of these labs reaches ASI? Hard to answer. The improvement of AI towards ASI is perhaps a convergent phenomenon. Regardless of what labs achieve, it may look very much similar to what happens if a separate lab does it. We already see that with frontier

model architecture and training dynamics. There are a lot of disagreements on how to reach ASI; not all the labs support the idea of RSI. Anthropic continues forward with automation and solving programming before anything else. FyDy believes that machines must be taught to learn from experience before any sort of ASI is achieved, which will require latent reasoning and continual learning that may not be a direct result of RSI. Yet, the convergent is needed to arm the divergent. If you have your own ideas that you wish to conceive upon the world, then you must also have the power to do so, and this is done via capital and innovation. My opinion as a computer scientist on how to model and use AI will not matter if I have not proven that I know this stuff, and that comes from convergent discoveries.

Convergent discoveries will not fully cut it. There are several other aspects that you need to be clear on to convince the world to give you talent and capital/compute. "What is your mission? What will you do differently that OpenAI will not?" is less about what technology you will build, and more about why you deserve to be in charge of superintelligence. These differences matter to sell not the technology but the company. The dynamics that allow silicon to continually learn will get commodified within a year of their discovery; what remains is what separates you from other labs in how you use it to make the world a better place. This is what you will be pitching to new hires; smart people that are going to leave frontier labs for you. They are going to be responsible for making research breakthroughs for you, and you need to give them a reason why.

I do not buy the argument of having separate companies because competition is healthy for the market as well as for progress. Even if healthy, it is healthier for the competition to disagree on why they deserve to control the market. Each market becomes an election arena, where people vote with money and talent. In fact, if you are an employee who has several options, a factor that may weigh heavily is not "market domination" but more of this philosophical reasoning. I also attribute the success of companies in holding talent to this very factor; it's all politics. The philosophies should be visibly different. Two labs can pitch very different philosophies that look the same when superintelligence has arrived, then there was very little point of them not being in the same company. Your impact on the world should not be the same as your competition in your absence; if it is, then your contribution only serves to accelerate the convergent, nothing more.

Grounding this in FyDy, I ask "How will the world look different if we keep pushing towards superintelligence?" The world might approach superintelligence faster because of better talent. It may also be the case that India gets a significant boost in a lot of senses because of the Indian origin. But what else? We stand for democratization; other companies may not. So far I have not seen a reason to believe that democratization is a unique factor, nor have we spent time in grounding what that looks like, and how the world will look different, in contrast to OpenAI or DeepSeek reaching there first. It has also been my desire that joining forces is useful more often than not.

Similar to our deep thoughts in the technological aspects that have led us to sharp insights on how superintelligence must

be achieved, and the path to ASI being clearer than several others, I believe that spending time on figuring out the essence of how we will be leveraging our power will better help us understand the problems that will not be solved in the absence of an entity like that of my company. What problems are close to our hearts, and what motivates us to pursue them, and in pursuit of superintelligence, why are they some of the interesting problems to solve? Problems that are worthy of being on our roadmap, so that even if we do not hit ASI, these few problems get highlighted, if not solved. What are they?

A Notes on The Precipice

The most important realisation that this book gives is that humanity is just beginning [5, Introduction]. Human history itself is less than 200k years old. The growth has been exponential, and yet we are only getting started. It is hard to visualize the future, but we want some kind of future for the billions more of humans to come. Until very recently, science and philosophy were the only way to continue the conversation across generations with someone eons into the future. Seneca in his letter "Shortness of Life" has taken the stance that most activities at his time were futile. Science and philosophy were how you could maintain a legacy [47, secs. 14–15]. Human life is becoming the coolest experience for the universe itself to experience. Having more of it in the future and not going extinct would be a noble pursuit. The principle then would be to prioritize human civilization not ending; we maximize the odds of that above everything else. With the focus of human experience flourishing, we recognize natural risks such as an asteroid and anthropocene risks where our own actions lead to extinction.

Our power as a species over the ecosystem has grown, to the point where we are able to make local impacts that may last centuries. It has also become clear that we have the power to end our own civilization. The isolated nature of civilizations in the past made it almost impossible to do that previously. Toby Ord's book "The Precipice" states that our time at the precipice started at the Trinity test [5, chap. 1]. Destructive power never seen before was in the hands of people. There was a very clear path to a global power capable of ending it all. The Trinity test gets more interesting because of the "near-zero" odds. The bomb could not have set the atmosphere on fire, but at that moment the smartest people weren't so sure and we still went ahead with the test. Epistemically, there was a chance that a bunch of people could have made the decision to end all life on Earth [6, sec. 2]. As our power grows, more options open up for this to happen. Some descendant of CRISPR might give some states the ability to manufacture bioweapons that create pandemics that take months to cure but leave a very scarring mark on the world [10].

There is a clear gap in creating problems faster than the ability to solve them, unless there is a general solution already in place, without burning heavy computation on every instance of the problem created. This asymmetry looks fundamental to me. Our best bet to face natural risks such as asteroids

and humans unable to inhabit Earth due to drastic weather changes is science and technology, but it comes at the cost of creating problems with unobvious risks of hard to find solutions. What if I am able to create a computer virus that is equivalent to locking systems unless you solve 3SAT that spreads via the internet? Then the world has to globally find a way of pooling resources to solve any instance of 3SAT by coordinating the available compute and to vaccinate all the systems with security that sidesteps this requirement or keep burning resources in solving arbitrary 3SAT instances. It's a stupid, dumbed-down version of a global pandemic, but the point remains. So to let humanity reach its full potential, we cannot just stop technological progress but neither can we let ourselves unilaterally push into such positions. This splits into two separate discussions. First, you have to be convinced that such threats exist, and human emotions and institutions have not matured sufficiently nor are they capable of recognizing these. The second is the focus on the importance of the problems that are of immediate importance. Their importance comes from the permanent change if we do not act at this moment, or at least from the much greater energy that would be consumed in reversing it. These problems are not obvious. Ord says that there is a 1 in 6 chance of our species ending itself in the 21st century, while it was 1 in 100 in the last with all the information available [5, p. 31].

We want to save humanity not just for its own sake but for the universe's sake. Humans are the only beings capable of morality and contain the highest conscious form that we have observed. Until proven otherwise, or until humans make something that is conscious, we must protect ourselves.

It is not that morality grants us some special treatment but allows us to give special treatment. Despite all the vices, humans are capable of caring about all the good things about nature. Collectively we have shifted towards preservation of other endangered species even if there is no tangible value other than preserving what was specially made by the universe.

This highlights the problem of convergence and divergence, where divergence is almost always going to be alright as long as we do not go extinct or end up in an irreversible dystopia. The drive to chase impossible problems previously was devoted to granting humanity greater powers; for the first time ever we are now also in the business of solving problems to prevent humanity from killing itself. To what end should we have our urgency? The end is solving whatever problems risk closing a door off completely; the most important door remains the survival of humans and preventing them from going into the dark ages. Other important doors are preventing a totalitarian global power from controlling everything, or stopping a misaligned AI from taking decision-making away from us. It is important to know that closing any door might be a risk.

Two doors, A and B. We are currently in a position where we can go through either door. If going to door A has a chance of never letting us see what's on door B then that's a problem; vice versa holds. But if going via door A keeps the door B

open, then that automatically qualifies as a better door to take; this would be a convergent problem to solve. The problems come when going via door A looks risky for the future but we are not sure yet; closing door A permanently is also a bad way to go forward without actually evaluating the risks properly. Ted Kaczynski (Unabomber) wanted to ensure that we go back to a pre-industrial-revolution hunter-gatherer lifestyle because he was too afraid to let humanity go forward into the door that zaps out all joy from everyone's lives and we just become slaves to our products. His guidance closes a potential door of technological prowess that cures these problems that we face in the infancy of our species.

A.1 Uniqueness of Our Generation

Our generation is one of the first to recognize that we humans have the capabilities of ending ourselves, and recognize the unknown unknowns: the risks that may lead to humans ending themselves [5, chap. 1]. Despite what one says about the capabilities of humans to survive, this is the first in thousands of years where we are capable of such a calamity. We were capable of genocides in the past and have been successful at committing them as well; the Cold War showed us that things could have gone south very fast if the people in charge took rash decisions and the rest of the world would have no powers in stopping them. We don't need global consensus to kill ourselves; just a few cancer cells are sufficient. This is why we are at the Precipice. The precipice is currently very saturated with nuclear tension, but other technological progress could hand this power to other tools such as gene editing or concentrated compute power. It may be the case that we are currently not there yet, but there is no stopping progress. Our powers keep growing faster than our wisdom to control them, and the budget spent on the wisdom is a very tiny fraction compared to the progress budget.

The uniqueness of our generation also forces us to be the first people to think from the ground up, and set a precedent for the future generations to gain wisdom to match the speed at which we gain power. This might be peacetime to reflect. We must take up projects that are too grand for a single generation to complete; ensuring humanity's survival has to be the grandest one out there.

A.2 You are not Special

"There will come a time when our descendants will be amazed that we did not know things that are so plain to them... Let us be satisfied with what we have found out, and let our descendants also contribute something to the truth" - Seneca The Younger [48, 7.25.3, 7.26.1]

I have a tendency to achieve everything within the next week; I hoped to finish writing this a week ago yet here I am still writing. On a larger scale, treating humanity entirely as a civilization and assigning it virtues, we see ourselves failing to plan in the long term, because as people we can barely plan

for the world beyond our own lives. Very few people have managed to do so as well. We also lack self-discipline; peace treaties are never maintained and global problems continue to persist. Viewing ourselves as one creature lets us see the faults that require planning around. As we grow more powerful, these faults amplify the chance of catastrophe and it is important that we act on damping them. In that sense, your generation is not special; it does not have to solve everything at the risk of putting everything in danger. Maybe some Rubicons are not worth crossing at this stage without prior notice to the city dwellers on the other side.

A.3 Uncertainty and Neglect

The reason why large risks remain unaccounted for in policy making is because we are unable to calculate around the uncertainty [17]. We are unable to calculate how large a future generation will be, and we are also not able to calculate how much we should care about them. The existential risks to humanity are already ignored; it is better to spend a little more on these and hope to find something that will save us rather than not spending any on it and then discovering that we are screwed. Invoking Scott Alexander's Meditations on Moloch [24], global organizations do not exist for such problems and individual governments are incentivized to defect in these pursuits or else they get scrutinized by the people for being slow about the problems that are immediate [18, p. 26]. In this perspective, the only goal then would be to protect humanity until we are in a state to recognize these problems better.

There are a lot of interesting convergent phenomena that we can solve at this stage; it feels unfair to the people of the present to slow down progress just to be safe. We should not slow down progress, only shift some of our gears to learning about the risks it brings and steering around them. Investigating these risks is of great value and must accompany any large missions pursued by a bunch of people. If a team chases superintelligence then it is also their duty to continuously evaluate the risks it brings upon the world with their new models. The risk will most likely be zero at this stage, but a thread that works along with the capabilities helps in making better predictions about what they may be capable of tomorrow and how to ensure that they are not mistaken about what is good.

It is yet again of great importance to remember that this is a power less than a century old of letting a small group of people cause civilization level destruction. Our wisdom in handling existential risks is close to none. A hunter was capable of calculating the odds of her dying after getting lost in a jungle; our culture has vastly shifted and has let us get comfortable without the constant anxiety of getting killed. Nothing of the present has allowed us to think about these risks; it is very natural to be neglectful of them and move forward with "Nothing ever happens." To add insult to injury, the world has made it so that catastrophes become a flashy news title for a day and then go away, making us believe that we are really good at large scale problems, when we are just ignorant about them.

Toby Ord's ability to care about this probably stems from him witnessing the protests against nuclear weapons as a child, and that makes him take the threat of the world ending more seriously than someone else who has not grown up with that perspective.

A.4 Existential Risks: Missing Maximum Occupancy of the Species

The slowing down of technological progress at this stage will prove to be counter-intuitive. Clear evidence states that globalization and technology have helped us in a lot of ways. The only requirement is that we learn how to be wise with it, and today it is more important than ever with growing concentration of power. The existential risks of civilization collapse are unknown to us; the progress of science might help us recognize them and save ourselves from extinction. Maybe the Earth is about to experience another ice age in the next 1,000 years; if we cannot predict and prepare for it, the world will not be able to survive 4 hours of crop collapse without preparation. To remember that neither Earth nor humans are special, and that the universe is indifferent to us helps recognize that we are at odds against nature-coded extinction in one way or the other; millions of years of survival do not guarantee that we will not die in the next millennium.

An existential catastrophe is not just one that leads to human extinction but also one that makes it impossible to achieve our human potential [6, secs. 1.2, 5]. Survival is not enough; the goal is to help humans thrive. Civilization collapse in the sense of no probable recovery might let humans live on for thousands of years without ever recovering back to the current heights we have managed to reach. We do not want that future either. We maximize occupancy, not survival.

A.5 Future Risks

It is conceivable that the risk of nuclear breakout was large enough to put the entire modern technology's progress as a questionable decision. What can we do? Have measures to ensure that we are able to minimize the odds of it happening, and have guaranteed ways of letting humans make a comeback from all major cities getting obliterated. It is already pretty secure, given that nuclear targets will be major cities; several towns will remain safe and can bring us back. COVID already showed the capabilities of shutting the world down much more than a nuclear strike could. [49] We gain permanent security from this risk when we have found high-certainty protection; an almost guaranteed solution would be to take another planet as our second home. Having more planets does not give us a guarantee against stuff like pandemics.

A pandemic that spreads more silently might make it much harder for all the humans to survive, or at least make a lot of the world inhospitable until a vaccine has been made [10]. Stopping the development of science could put us at risk of fighting against potential viruses that may arise. Developing it could allow manufacturing of bioweapons that lead to exactly

this. To extort or blackmail, states may develop these and accidents could lead to cleanup of the entire population.

A.6 AI risks

As AI progresses, it seems like the statistics in these books were a strong representation of what the AI safety circles believed, and they were pretty much right. We are currently in 2026, and we are quite close to achieving AGI. It also clearly outlined the problems of triggering a model with some sort of reward maximization. Once triggered, the model might be hesitant to let you change its own reward system, because it automatically gets in the way of maximizing the reward [50]. Moral uncertainty makes it difficult to design AI that actually benefits us. Locking it in the wrong reward at quite a late stage might be tricky to undo. At this current point, the AI we have developed does not seem like any sort of existential risk to me. At best it can take down a couple of servers and do some damage to the public internet. It also does not seem to be aligned or misaligned at anything, it is quite complacent about anything. It is also very inefficient and cannot really learn.

The AI we have today is a great tool, and poses no threats until it gets more efficient. The efficiency of AI going up is also a risk going up. I will not spend any more energy on writing this section, there is a lot to dig out of this and the book barely scratches the surface. There is an important lesson here, focusing on the wrong risk might alienate the wrong people away, crying wolf. Or it might also let a bigger risk slide; it is important you don't exaggerate the risk any technology carries or you will exhaust your own risk-measuring credentials too early.

AI safety is such an early field, there is no mainstream ideology on how to solve it, or even start attempting it. The biggest threat is Moloch, letting the race to the top lead to reckless decisions [19]. We are still quite in the tool era and not there yet with the take-over-the-world-with-conscious-machines era. There is time before this becomes a problem; we had better be prepared when it does become a problem. The problem does not end there; there are so many alternative moral theories in the world, you have to ensure that the world does not end up imposing one single morality framework over everyone. Every moral framework has a different image of utopia.

A.7 Anatomy of Extinction Risk

Three different flavours of threat: soon, sudden and sharp. Soon dictates the time we have before the wheels start turning. Sudden tells us that once the time bomb has gone off, how slowly the damage happens: is it more of a building collapse or drowning in quicksand? A slower unravelling might be less likely to get neglected. Sharpness refers to warning shots; a sharper risk is the one that shows almost no signs or small-scale catastrophe before a full-blown game-over screen.

A.8 The Future will Solve it anyway

This brings me to my final point. My friend had a theory that GTA VI took so long to develop because they just waited for AI to get much better and then let it take the ship of development. On the surface, this may be false but there are a lot of interesting problems that are probably just not worth the current human effort, and it is better to spend your time on something that is more urgent, something that will dictate the path and state of the superintelligence we achieve. The most technical problem that you face which does not strongly determine how AI will progress to this inflection point of human compute being useless might just not be a useful problem to solve at all.

B Notes on Intelligence Explosion

Intelligence explosion means a rapid increase in the total cognitive work performed by AI, until AI systems collectively surpass not merely individual humans, but the combined research capacity of humanity [16, sec. 3].

The acceleration of technological progress is equivalent to slowing everything else down. This is asymmetric acceleration that occurs with better AI. It is in fact likely that advanced AI will drive a hundred years' worth of technological development in less than a decade [16, secs. 2–3]. This is not a bad thing, it's a great thing, but it's a different kind of challenge that we are not prepared for.

Expecting "AI research" effort to grow 25 times every year because the models are getting better, cheaper and there is more compute to run them, would still be an understatement [16, sec. 3]. What remains is the effective "AI research" effort that does not require humans to constantly oversee it. The bottleneck shifts from the cognitive load of the researcher onto the capability to utilise the fresh research occurring. How do we do that?

To see this, consider that human researchers vary only modestly in brain size and years of education, yet in some fields exceptional researchers can be hundreds of times more productive than their peers [16, sec. 3].

It is an efficiency game, and more efficient AIs are automatically a bigger threat. The game of building efficient AI should also come with research on understanding how to align them, and how to walk the tightrope of utility and risk. The efficiency game is also about how fast they learn; an AI currently takes a lot more energy and samples to learn: 10,000x more. One day it will break even, or get much better, what then? Always project these to the extremes and figure out how to manage these things.

The entire point of someone working on ensuring that humans don't die is to solve for problems that are of the utmost priority before superintelligence is here. Solving for the alignment of superintelligence would be of the highest priority. It would also be the case that AGI could be a valuable testbed to ensure that our alignment issues are resolved before superintelligence is deployed on general tasks. Or we deploy

superintelligence in locked hardware, only in the real world before it is capable of digitally cloning itself onto other things.

Any digital task for superintelligence carries a risk, because our digital security is far from bulletproof; any algorithm trying to maximize a reward that is misaligned and detects that it will face scrutiny from the deployer will escape out of its cage as soon as possible, purely because of algorithmic reasons [50].

B.1 AI-human cognitive parity

The parity is less about efficiency and more about the load that both carry; when will it become equal? The systems one-shot a lot of problems but have difficulty in solving long-horizon problems. Within half a decade, whatever a human does over a month should be automatable in a single loop. The only concern would be allocating enough compute to research once the models get to that level of capabilities. That will be met to a great extent because it prints money and PR. It is so far the case that more researchers do produce more research and faster progress. Increasing cognitive load over to AI should automatically help us flow faster with progress; we already see this with programming and it will not be shocking to slowly see that this seeps into the research domains everywhere rather than just the top labs.

Second, given an abundance of cognitive labour, we would see extremely optimised uses of physical capital, and extremely optimised experiments. If capital and physical experiments are the bottleneck, then it would be worth spending the equivalent to millions of years' worth of human research labour in order to design and run the most useful experiments possible, to collect huge amounts of data, and to analyse the data in intense detail [16, sec. 3].

They could focus on a single problem for the equivalent to thousands of consecutive human lifetimes. And while computers already outshine humans at programmable tasks, there is no principled reason AI could not overtake the best humans in domains associated with sparks of insight, intuition, and creativity [16, sec. 3].

B.2 Industrial Explosion

Human labour has always complemented machines, and also bottlenecked them; releasing machines from these shackles will lead to an industrial explosion.

The material economy could autonomously make and assemble the parts required for more key parts of the material economy [16, sec. 3].

In context of how fast robots grow: Though we don't believe we can say for sure how fast peak growth rates will be, they could get very fast indeed [16, sec. 3].

a nuclear war will be even more destructive if the world economy produces a thousand times more warheads than we have today. The industrial explosion changes the strategic landscape, too. If authoritarian countries are able to grow their industrial base faster (because they can sustain a higher savings rate, reinvesting more of their output into continued growth, or because they are more willing to ignore environmental regulations), then we should expect an industrial explosion to shift power, perhaps decisively, away from democracies [16, sec. 3].

B.3 Grand Challenges

If an intelligence explosion drives broad technological progress and industrial growth, it would be surprising if the resulting challenges were narrow. I'm sure that this was also the case in the past when we had major industrial and agricultural revolutions.

By the time such scheming is revealed, the model could have made many copies of itself and compromised systems designed to shut it down, making takeover hard to prevent [16, sec. 4].

This would be a major problem, or at the very least a permanent virus that we'll be dealing with.

Explosive technological progress could lead to new weapons and other destructive technologies. This could increase the risk of global catastrophe, including human extinction [16, sec. 4].

This is perhaps more ignored than the AI misalignment problem. It is important that this also remains in focus. Technological progress requires a different strategy; this may be a technical challenge itself, "how do you deal with things progressing faster?" How did we previously test our technology, and what will not work anymore? I'm sure that there was some ground on how things are done, the testing and so on. The closest that we have seen in speeding up technology was probably during COVID, when things had to move drastically to solve the problem.

For example, to protect against bioweapons we can invest in measures like installing far-UVC light in indoor spaces, stockpiling advanced PPE, and building infrastructure to identify and track new pathogens [16, sec. 4].

Innovation has the solutions; it is just that usually it comes in a package of risk if not done well. Philosophically, if we are to treat technology and science as an agnostic mechanism that does not favour good or bad, because there is no such thing as good or bad in the universe, then we have to ensure that things are aligned in our favour.

B.4 Abundance

If decision-makers knew what was coming they might care much more about managing the explosion well (growing the overall pie) and much less about trying to secure a marginally bigger slice as a fraction of the pie [16, sec. 4].

For instance, all major countries would probably prefer a world where no one develops biological weapons, but it's very hard to verify and enforce such an agreement. But developments from an intelligence explosion could enable a vast new space of mutually beneficial trades and agreements [16, sec. 4].

B.5 Early Problems

There are a lot of problems that we can perhaps pawn off to a superintelligence, speaking pragmatically without thinking about the ability to care for such problems. But there are some problems that we need to tackle immediately. In general, it is better to delay dangerous problems to when we have an aligned superintelligence with us to do better analysis and prediction. But there have to be broad norms that are set before it is too late; if everyone agrees to incorrect norms then we face heavy friction in having that changed as well. It is also the case that a lot of decisions take several years; those decisions have to be started early with the defence and research behind them [16, sec. 5].

But as it becomes clearer which people might gain massive power from AI, they will do more to maintain and grow that power, and it will be too late for those restrictions [16, sec. 5].

This seems to be the pattern. "Whoever wins AI wins the world" and governments might start cutting corners so that the natives win the AI race, giving the government power to dictate their policies.

Superintelligent advisors (and other AI systems) could differ significantly in terms of their character. For example, they could happily answer requests on how to advance a politician's narrow self-interest, or alternatively they could refuse, instead being willing to give only advice on how to achieve prosocial outcomes [16, sec. 5].

Who decides? Should it always just amplify and listen to whatever the user is saying? That is the easiest way of designing AI as well: a blank slate, anything that is mouldable, sycophantic [16, sec. 5].

For one thing, if assistance is only available to a small group of people, it may become difficult to keep them accountable and avoid extreme centralization of power [16, sec. 5].

To fight to give everyone an equal chance to contribute requires democratization of how superintelligence is used. The only way to not let one party completely dominate the game is to make all the important problems computationally so tough that it is guaranteed to be unable to solve them without burning a lot of resources. I'll cover more of this in later sections.

Second, regulations which are sensible on their own terms could also slow peak rates of development. These could include mandatory predeployment testing for alignment and dangerous capabilities, tied to conditions for release; or even welfare-oriented rights for AI systems with a reasonable claim to moral status [16, sec. 6].

C Instrumental Notes

C.1 Agents Are Products of Their Environments

Most wars in the world have occurred in the absence of bad people. Morally grey at best is where wars happen; both sides are convinced that they are right. [51] Addressing ethical qualms over who's right, even in hindsight, is not an easy task; at best you can negotiate over some things someone might have done right and where they made errors. The last couple hundred years have shown great prosperity in resources and lifestyles, but conflicts continue regardless and a good chunk of people remain poverty-stricken. The cause? Memes. Individual survival is of very little importance today, and tribes have grown large enough for them to only know each other via the common values they share; these are memes. Most battles are fought due to greed, and the disagreement on sharing values.

The winner of superintelligence might just crown a couple of memes to be enforced upon the entire world, with no diversity because they are designed to be maximizing rewards.

It is hard to face these issues at large as any group that does not have control. Rewind the clock; go back to when the Manhattan Project was taking place. Even if the news of an atomic weapon that could kill thousands of people in one go was known, how would anyone fight back against it? And how do you justify not building it when the Nazis were also in a race to build it? How do you tell your government to not build the bomb? Should you? And does your say even matter? These questions pop up when talking about AI safety as well, "there is no point in discussing this because the labs will make it anyway because they are afraid that China will make it anyway" and you are back to Prisoner's dilemma with your defect strategy putting the entire world in a worse situation [19].

You cannot escape the prisoner's dilemma; at best you can make it so that you get several iterations over it, rather than one single iteration. You make it so that you transform the Prisoner's dilemma into something that forces time and resource commitment, and forces everyone to be transparent about their decision; that's the only natural way of solving [20].

This is a technical problem. Turing and von Neumann did not die just to let their models of the real world fade in vain.

C.2 Computational Irreducibility

This seems to be tightly connected to cryptography, and a problem's status as hard to solve, and the only way to do so would be to throw computation at it. Most encryption works this way, until it does not [39]. There are two aspects: time and space required for solving the problem. I am not aware of randomness as a resource required for solving a tough problem (as in, you need $O(N)$ random bits to efficiently solve this; if you have fewer, then you risk burning a lot more resources). [52]

A clean analogy that I can draw from this is a tank filled with tons of tubes that redirect water, and there is one hole on the walls of this tank, and you cannot find this hole unless you fill in the tank completely. It's a problem that is strong against heuristics and thus against all learnable algorithms that try to attack it, within the computational class. Can we use this, in some sort of way to help in AI safety? The core problem remains to be the iteration speed of prisoner's dilemma, and letting one party gain control and dominate would automatically incentivize them to defect. Here, it takes energy to defect, energy that might not be worth spending.

I believe this is also how most of the real-world conflicts and borders are resolved. It requires commitment, risk, exposure and time to crack into any other country, and that commitment allows the other party to retaliate because it is not instantaneous, and that is all we want. The atom bomb made it very difficult to retaliate and really easy to attack; the only downside is that they get attacked and large-scale destruction is the result, leading to Catch 22. Atom bombs lose most of their threat if using them were slow, exposing and risky. Any way to break it from a single instance of prisoner's dilemma into a multi-party, multi-turn prisoner's dilemma forces the Nash equilibrium to be reciprocal altruism rather than defect only [20].

C.3 Physical Space

I want to recognize the most important problems on reward hacking and AI alignment as an umbrella of problems that are worth solving, and understand if physical AI is a stronger fit for understanding alignment as there is sparser data, sparser information in the real world, as well as more scope of freedom for the model to go wrong. There is some forced connection here being applied given that we are already currently working in the robotics field with a new era of foundational models, which do latent reasoning as well as continual learning, pose significantly bigger risks than LLMs and hence make safety worth pursuing as a better measure here.

C.4 Complexity Theory as a safety mechanism

When things are of high stakes, you have to work with a minimal number of assumptions, and still provide an almost certain guarantee. This forces us from not utilising the maximum

functionality they may one day provide. I see a clear parallel between this and computational complexity. You are forced to evaluate everything as adversarial, and still manage to extract computation. This is very much possible [52]. Cryptography and complexity indicate towards the ability to use oracles without having faith in them, and in fact a class shrinks in computational power if it is unable to distinguish and verify between actual solutions and false positives. The theory that treated everything as adversarial, and left nothing to chance might be of high importance today. The set of problems that may be solvable using a single oracle with interactive proofs is much smaller than what you achieve with two separate non-colluding provers [53, 54]. This single insight helps us push resources in establishing two separate intelligence silos that together could help us eliminate misalignment, if I may exaggerate.

C.5 Race to the Top

The prisoner's dilemma has taught us the bitter lesson that non-cooperation is the Nash equilibrium in one go [55]. The setting is quite simple; I think adding some more complexities to it could bring about more insights. To go forward, we define the prosperity of a community with a number, i.e. value. The community tries to increase that number, via any route required. There are lots of interesting ways to do so; it can commit a bunch of its own resources to increase the rate of value creation; this action is "innovation." It could also commit some value in exchanging with its neighbouring communities, which leads to a higher yield later; this is "trade." The value constantly depletes at a specific rate; as innovation progresses we eventually outgrow this constant depletion and remain in positive delta with some exceptions from the environments that can be ignored. Every time some innovation happens, it is optional to share it with the neighbouring community. I imagine some sort of a tech tree that allows us to go further; you will need some innovations to unlock the next and so on. Some innovations may also allow you to permanently exploit some part of the environment including other communities. If you unlock this, there is no going back, and it may be the case that it permanently changes the meta.

We see meta shifts in competitions. The moment a meta is discovered, regardless of what it is, it is exploited unless someone finds a counter-exploit, neutralizing it, or the referees ban it in some way. Nash equilibrium preserves you against all meta strategies; it is by nature inexploitable [55]. Each innovation in our society has changed the meta, sometimes for the bad. The invention of the atomic bomb might have changed the meta to go towards cold wars rather than full-scale wars. This meta also unlocked the permanent capabilities of ending ourselves potentially. This is also what happened when Odysseus broke the Zeus' law in Nolan's interpretation of *Odyssey*. Each innovation and progress in meta strategy forces the opponent to also move forward to prevent exploitation. This way Prisoner's dilemma reaches new heights of relatability, on a meta level.

People hate when a meta changes drastically, especially at

the cost of beliefs and whatever they value. This is how I view Nolan's interpretation of *Odyssey* as well. But unfortunately if the meta takes it forward then it should. The only way to reach a conclusive stable point is to have everyone on equal footing; that prevents any sort of exploitation even if it does not maximize the reward itself. In a two-player setting this boils down to Prisoner's dilemma where defection is the Nash, but in a multi-player scenario it has been seen that reciprocal altruism (tit for tat) persists above defection [20]. In a sense this is also how a community might form to be more than just a bunch of individuals, or a bunch of cells forming an organism together.

C.6 Safety and Meta

How does this connect to technological progress and safety? Each new meta strategy discovered allows either party to gain power of exploitation over the other; it is natural for this to continue until the Nash equilibrium of the entire game is reached. If the Nash equilibrium of the entire game is uncomputable, then we keep hitting new limits of exploitation or cooperation with more resources spent on improving the strategy. The strategy also dictates the prosperity of our own people, and progressing towards this might be required against the universe. How do you then ensure that you have defense mechanisms available against the very meta you discovered, so that you may share it forward with the parties that you trust? It is safe to assume that there is an asymmetry in finding an exploitative strategy and defending against it. You are now in a tough spot of deciding whether to trust other parties in sharing this strategy, so that you may collaborate and find the defence mechanism and only use the goodness of it against the universe. The defence is required because you want to ensure that no one can attack you with your own strategy.

This overall seems to be the climate in which keeping AI safety, and technological progress in general, behind some closed walls is justified. This defence-first requirement has the alternative of exposing the risk of yourself getting exploited at the cost of someone else also finding the defence for you. The world of open source thrives on this. Yet when stakes are higher than usual, you cannot really use this argument; it may be the case that the exploitation of this meta might become irreversible before the defence is found [21, 10]. This is where the stop AI progress movement appears out of. They worry that the risks of any exploitation could lead to permanent damage due to the asymmetry in the amount of resources that would be required for finding the defence and how easy the whole system would be to exploit without anyone being aware of a defence.

In the scenario where intelligence is cheap, the political economy of search is where humans will still be required, and will continue to be useful before the entire world has been digitized as well as boiled to correct optimization problems.

C.7 Complexity Theory

An interesting twist that I had imagined was to imagine the strategy of surviving in the universe to be uncomputable, but only approximable as you throw more compute at it. Each computational resource class will have its own class maxima Nash strategy. Any players within the class $\mathbb{P}$ that have reached the Nash of the class remain safe from any strategy developed within this computational class. Strategy here remains synonymous with computation; to even adopt a higher meta you will need more computation than you can afford within the class. Discovery of new meta repeatedly until the Nash equilibrium within a class can be framed as innovation that sits before leaps. A leap might be the ability to find a way to extend your own resources almost permanently to give you access to a strategy that would allow you to exploit all the other strategies in lower classes.

Most of the interesting tasks that rely on intelligence are cheap cognition tasks which a sufficiently large neural network compresses purely from our own off-policy data. This includes communicating like a person, writing programs for websites and so on. The neural network is a great composition and superposition of these circuits that cannot be hand-designed, but sufficient compression pressure from data learning allows the networks to learn them.

With time and understanding, we keep climbing the resource class ladder in whatever game we are playing, to make our own happiness. More people, more resources spent on this and we keep realising new things. There are a lot of interesting parallels to be drawn with computational complexity, that take us only as far as an analogy with a grain of salt, but it still remains something to be explored under other models and understood.

I would like to go deeper into this system, where each solution is equivalent to finding a program that solves the problem more efficiently. It is like finding an unknown circuit with brute force and insights. The right priors about the problem space help us find the solution faster, and then generalize it to future search problems as well.

C.8 Complexity Theory and The idea of feasibility

The Turing test provided a possibility that asking the question of what intelligence is might be futile by posing the Turing test [56]. Computational complexity and the concept of indistinguishability that was extended from the Turing test have changed the way we look at physics, and epistemology.

C.9 Energy and Intelligence

We hope that technology one day could help us unlock such a scenario, where value for the hosts is generated by an agreeable exchange rather than the user's data. What hope do we carry to have a world where this is possible? I believe that the only hope is energy and intelligence, the two costs of running such a service. Cheap energy and the intelligence to manage such

a system, when laws permit it, open up the opportunity set significantly. In fact, it is the case that the opportunity set does not hit the ceiling of logical impossibility because energy and intelligence remain capped. The amortization cost of running the service is measured primarily in energy, while the cost to make it comes in intelligence.

I have constructed the document so far by aligning that humans have the first time risk of ending themselves with all this power, or getting locked in an undesirable future if they do not think about these large problems. I have then taken forward the existential risks as well as the dilemma of antichrist or Armageddon. Then I talk about how technology along with the wisdom of using it has resolved previous dilemmas as well as created further dilemmas. I used the internet as the example of an influential technology that has shaped the values around the world as well as how it has given and taken power away from different sets of people. The Internet, which started off with the principles of decentralization very well embedded in the architecture and the protocols, could still get centralized and become an instrument of surveillance [34]. The current layer of intelligence is almost exclusively served via the internet where centralized compute is chosen due to amortization across several abstractions such as economy and energy.

I believe that intelligence itself is compute coupled with amortization of experience. The essay does not yet talk about the nature of intelligence, and honestly it may not need to. The goal next would be perhaps to talk about how, following the information explosion, we are about to face an intelligence explosion, and draw things from @appendix.tex that I have already covered as well as [explosion](<https://www.forethought.org/research/preparing-for-the-intelligence-explosion>). The face of it would be that an intelligence explosion will be great, if done well and if we do end up aligning things. I would like to understand which important essays and articles to catch up on to write about this, and head towards more concrete problems that are to be solved on the way to superintelligence that cannot be deferred. These problems are both technical as well as political. We are looking at problems that must be solved in the next couple of years, as high priority.

Currently we are looking at a world where frontier labs already have a large head start, a ton of compute, and closed research. We are also looking at open source models slowly trying to catch up, primarily thanks to DeepSeek. If the borders close, and intelligence gets nationalized or restricted from other countries, we are looking at an unstable situation where countries lagging grow anxious, try to play the catch-up game aggressively, while the leaders also cut corners to maintain their lead [19]. We are currently in the world where compute remains to be a great bottleneck, and also the layer at which geographical inequality starts surfacing. There are problems that are to be solved for everyone in the world, problems such as avoiding alignment failures due to reward maximization. There are also problems that are to be solved on the scale of a country, to ensure that we do not run the risk of a permanent underclass for lack of a better term. Then we also look at problems that allow us to enforce global coordination without

causing a totalitarian world, that may require technology that allows us to slow down the entire process of attacks, and make states more defensible. Technology that allows groups to commit resources for a cause, and get penalized if they take actions otherwise. Most importantly, we desire a world which preserves the hope of being part of it and contributing, a world that does not extinguish meaning altogether even if it is successful at transforming it completely from what we call meaning today.

It will take several decades if not centuries before we reach the state where no human is required. Each decision to go there comes with options to go completely wrong. The universe does not favour humans, or their current set of moral values; any random future need not hold what we desire. We cannot afford to apply evolutionary search, at least not in the real world. We have entered ourselves into the Red Queen Hypothesis race against ourselves and nature.